House Price Prediction

Project Overview

The objective of this project was to develop a machine learning model capable of predicting house prices using various property characteristics such as area, number of bedrooms, bathrooms, parking availability, and other amenities. Accurate house price prediction can help buyers, sellers, and real estate businesses make informed pricing decisions and estimate property values more effectively.

Dataset

The Housing Prices Dataset obtained from Kaggle was used for this project. The dataset contains information about residential properties along with their corresponding selling prices. The data was explored, cleaned, checked for missing values and duplicates, and categorical variables were converted into numerical form using encoding techniques.

Methodology

The dataset was preprocessed and divided into training and testing sets using an 80:20 split. Two machine learning regression models were trained and evaluated:

1. Linear Regression
2. Random Forest Regressor

The performance of both models was measured using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score.

Results

Linear Regression

- MAE: 970,043.40
- RMSE: 1,324,506.96
- R^2 Score: 0.653

Random Forest Regressor

- MAE: 1,021,546.04

- RMSE: 1,400,565.97
- R^2 Score: 0.612

The Linear Regression model achieved better performance than the Random Forest Regressor. It produced lower prediction errors and a higher R^2 score, explaining approximately 65.3% of the variation in house prices.

Key Findings

The correlation analysis indicated that features such as property area, number of bathrooms, bedrooms, and parking spaces had a strong influence on house prices. Larger properties generally had higher prices, and houses with additional amenities tended to be more expensive. The house price distribution was positively skewed, indicating the presence of several high-value properties. An interesting observation was that the simpler Linear Regression model outperformed the more complex Random Forest model on this dataset.

Business Recommendation

Real estate businesses should focus on key property characteristics such as area, bathrooms, and parking facilities when estimating property values. Implementing data-driven pricing models can improve valuation accuracy and support better decision-making for both buyers and sellers.

Conclusion

This project successfully demonstrated the use of machine learning techniques for house price prediction. Both models provided reasonable predictions; however, Linear Regression achieved the best overall performance with an R^2 score of 0.653. The results show that property features can effectively be used to estimate house prices and support real estate market analysis.